Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

9/1/2026

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session	**2024-25 (EVEN)**	**Course Name**	**Mobile and Web Application for IOT Lab**
Semester	**6**	**Course Code**	**23IOT1624**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**02**
Course Outcome	1. Understand and apply basic Kotlin programming constructs including variables, control structures, functions, null safety, and object-oriented concepts for Android app development. 2. Design and develop responsive user interfaces using XML layout files, ViewGroups, and widgets in Android Studio, adhering to Material Design principles. 3. Implement event handling and application logic using Kotlin to capture user input, manage app state, navigate between screens, and interact with Android system components like Toasts and Dialogs 4. Integrate learned concepts to develop a mini Android application that demonstrates use of ViewModel, multi-screen navigation, and proper UI/UX design for real-world problem-solving
Aim	Understand Kotlin functions, null safety, and OOP concepts
Problem Definition	Design and implement a Student Profile Management System using Kotlin that demonstrates functions, null safety, and object-oriented programming concepts.
Theory (100 words)	➢ Functions in Kotlin are used to perform particular operations and help in dividing a program into smaller, reusable parts. Kotlin allows the use of default parameters so that functions can work even when some arguments are not provided. It also supports named arguments, which improve clarity by allowing values to be passed using parameter names. ➢ Kotlin provides nullable types to safely handle missing or unknown values. By adding the ? symbol, a variable can store null, preventing unexpected runtime errors. ➢ Null safety operators help manage null values effectively. The safe call operator ?. accesses a value only if it is not null, while the Elvis operator ?: supplies an alternative value when null is encountered. ➢ Classes define the structure and behavior of objects. A parent class such as Person can store common attributes like name and age and provide related functions. ➢ Inheritance enables a class to acquire properties and methods of another class. For example, a Student class can extend Person and include additional student-related details.

	➢ Constructors initialize objects during creation. The primary constructor sets basic attributes, while secondary constructors offer alternative initialization methods. ➢ Data classes simplify data handling by automatically generating methods like toString(), equals(), and hashCode(). ➢ Enum classes represent a limited set of constant values such as ACTIVE, INACTIVE, and GRADUATED, ensuring better readability and type safety.
Procedure and Execution (100 Words)	Step for Implementation: ➢ Install and configure the Kotlin development environment using tools such as Kotlin Playground, Android Studio, or IntelliJ IDEA. ➢ Write a function that prints student information using both default parameters and named arguments. ➢ Define the student's email as a nullable variable by using the ? symbol. ➢ Use null safety operators like the safe call operator ?. and the Elvis operator ?: to manage null values without causing errors. ➢ Design a parent class named Person containing common attributes and a method representing the role. ➢ Create a Student class that extends the Person class to reuse its properties and methods. ➢ Implement both primary and secondary constructors in the Student class for object initialization. ➢ Develop a data class StudentRecord to efficiently store student-related data. ➢ Define an enum class StudentStatus to represent different student states. ➢ Apply logical conditions to decide the student's status based on age. ➢ Output all student information to verify the correctness of the program.

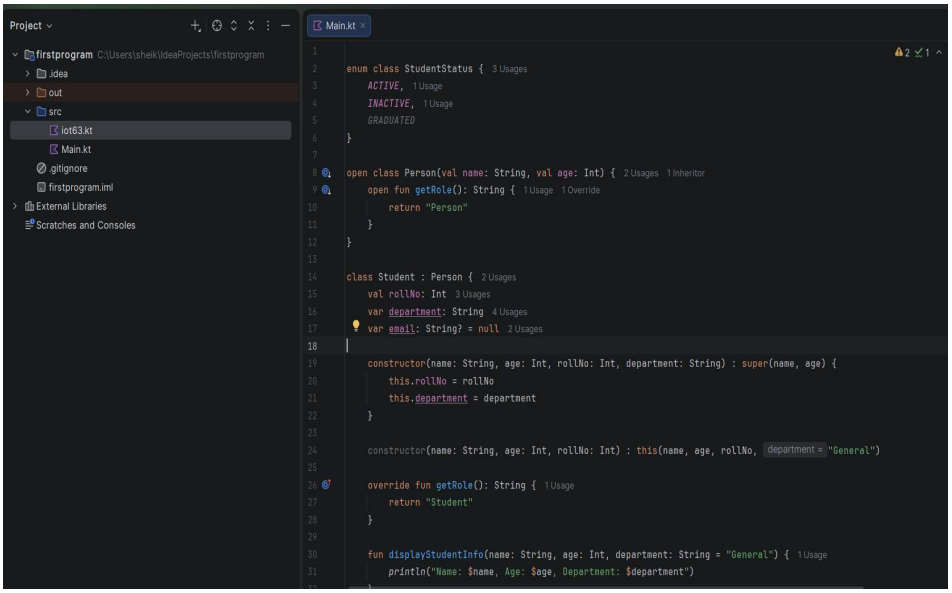
Vision of the Department

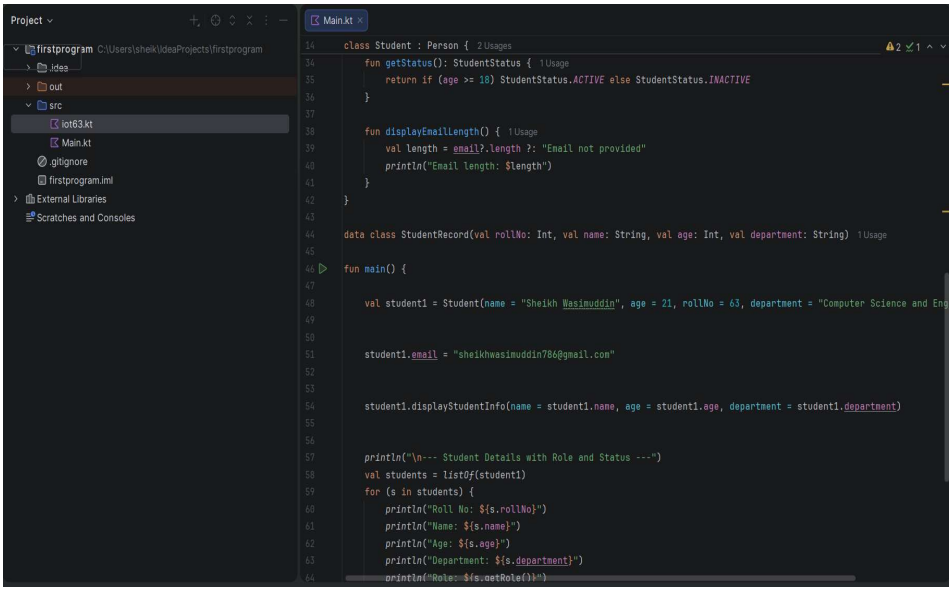
To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Code:



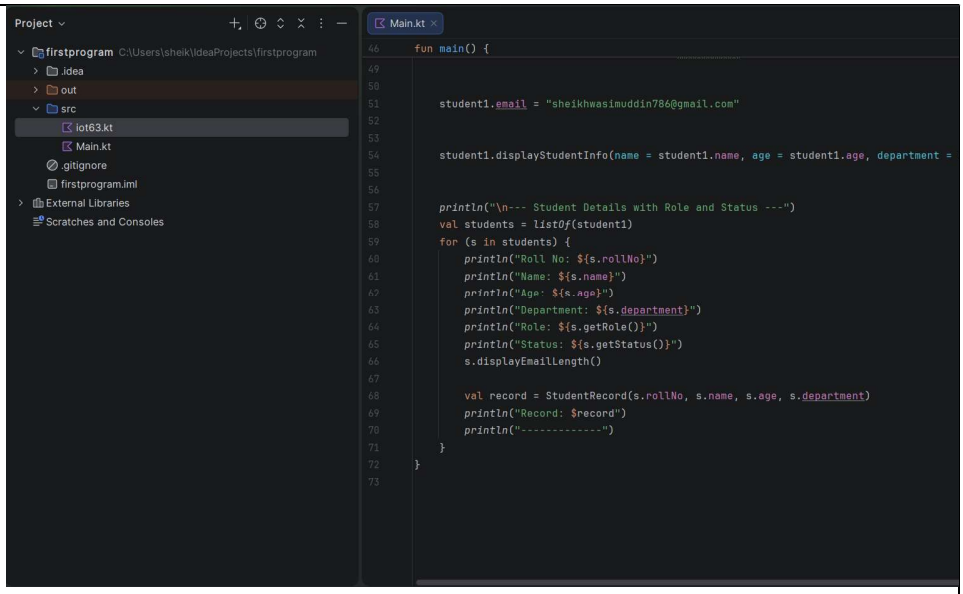


Vision of the Department

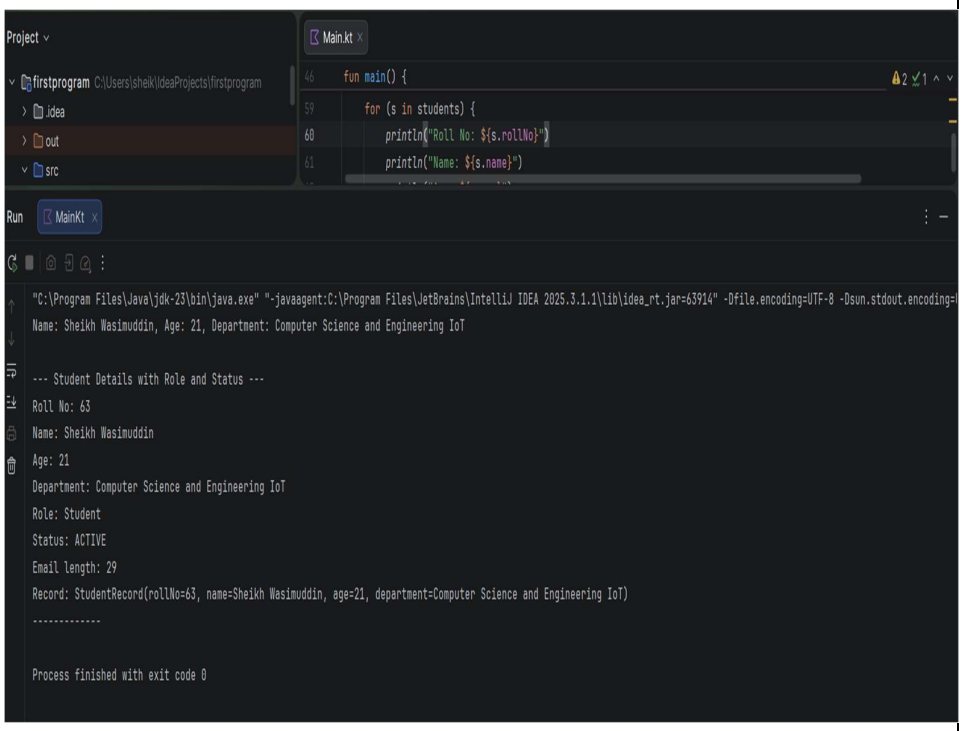
To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.



Output:



Output Analysis	The program executes successfully and displays complete student information, confirming the correct use of Kotlin concepts. It first prints the basic details such as name, age, and department using function arguments. Then it shows the detailed student record including roll number, name, age, and department, which are obtained from the Student class inherited from the Person class. The role method correctly identifies the person as a student, and the enum class assigns the status as ACTIVE based on the age condition. The nullable email variable is safely handled using null safety operators, and its length is displayed without causing any runtime error. Finally, the data class output prints the student record using the automatically generated toString() method, verifying proper implementation of data classes, inheritance, constructors, null safety, and control logic in Kotlin.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	The program successfully demonstrates the core features of the Kotlin programming language. It shows the effective use of functions with default and named arguments, nullable types with proper null safety handling, and control flow for decision making. Object-oriented concepts such as classes, inheritance, constructors, and methods are correctly implemented through the Person and Student classes. The use of data classes simplifies data management, while the enum class ensures type-safe representation of student status. Overall, the output confirms that the program is well-structured, error-free, and clearly illustrates how Kotlin supports safe, modular, and reusable code development.

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Plag Report (Similarity index < 12%)	SmallSEOTools **Plagiarism Detection Report by SmallSEOTOOLS** 4% Plagiarism 4% Partial Match 0% Exact Match 4% Unique 96% **Scan details** Total Words 1024 | Total Characters 7305 | Plagiarized Sentences 1.84 | Unique Sentences 44.16 (96%) **Plagiarism Results: (2)** **#1 2% Similar** https://www.scribd.com/document/867363264/138-... Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety. **#2 2% Similar** https://www.studocu.com/in/document/sharda-univ... Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and
Date	**09/01/2026**